BRAIN lab
basic research of artificial intelligence

LECTURE 8

Evaluating Language Models

Tasks, measurements and evidence

26 October 2026

From a question to a supported conclusion

01 What is evaluated: base models and assistants

02 Metrics: from exact match to rubrics

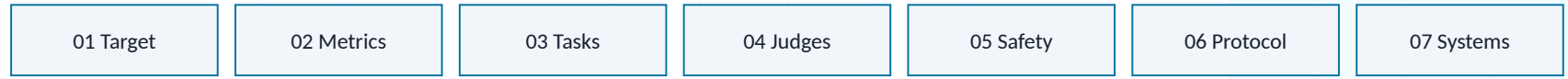
03 The benchmark landscape and coding evaluation

04 Judges and arenas: rubrics, preferences and ratings

05 Safety: harmful compliance and unnecessary refusals

06 Protocol, logs and reading reports

07 Evaluate a research agent; design your own test



01

What we evaluate

Turn a generated response into a checkable observation

01 Target | 02 Metrics | 03 Tasks | 04 Judges | 05 Safety | 06 Protocol | 07 Systems

How would you score these answers?

USER REQUEST: WHAT IS THE CAPITAL OF FRANCE?

Generated response	What should count?
Paris	Correct fact
Paris.	Same fact; different string
The city of Paris	Same fact; longer wording
Lyon	Wrong fact

First define success for the task; then choose a way to check it.

One dataset score starts with individual answers

Capital of...	Raw answer	Gold	Correct? (normalized)
France	Paris.	Paris	1
Italy	Rome	Rome	1
Japan	Kyoto	Tokyo	0
Germany	Berlin	Berlin	1

Accuracy = (1 + 1 + 0 + 1) / 4 = 75%

Benchmark = examples + task definition + scoring and execution protocol.

Base models and assistants answer different questions

	Base / pretrained	Instruct / assistant
Decision	Which pretraining checkpoint?	Which assistant behavior?
Protocol	Completion / option likelihood	Chat template + generation
Measure	Held-out loss + task scores	Task scores + instructions / safety

Evaluate the model in the interface and conditions your claim describes.

Qwen, 2025 · Qwen3-8B · thinking and non-thinking modes

ILLUSTRATIVE VALUES

The same question can use two evaluation protocols

QUESTION: 2 + 3 = ? **A. 4** **B. 5** **C. 6**

Score the options

log P(A) = −2.0
log P(B) = −0.3 → choose B
log P(C) = −3.0

Generate an answer

Output: “five”
Extract / map: five → B
Compare B to the gold answer

Generation needs answer handling; likelihood needs a specified scoring rule.

02

Metrics

Choose a metric that preserves the meaning of success

Three ways to check a generated answer

Known answer

Gold / reference available.
Compare text, meaning or mathematical value.

Runnable check

Run code or actions.
Check tests and the resulting environment.

Explicit criteria

Rubric = criteria + scoring rules.
Example: every claim must cite supporting evidence.

State what is scored, how it is checked and what gets averaged.

Exact match: specify what “the same” means

Gold: Paris	Strict EM	Our normalized EM
Paris	1	1
Paris.	0	1
The city of Paris	0	0
Lyon	0	0

Mean strict EM: 1/4 = 25% · Mean normalized EM: 2/4 = 50%

Accuracy averages correctness; the checker defines what “correct” means.

Rajpurkar et al., 2016 · SQuAD: answer exact match and token F1

Token F1 gives partial credit for answer overlap

CLASSIC EXTRACTIVE QA METRIC · STILL USED IN QA BENCHMARKS

Reference

New York City

Prediction

New York

Precision P	Recall R	F1
Overlap / predicted tokens 2 / 2 = 1	Overlap / reference tokens 2 / 3 ≈ 0.67	2PR / (P + R) = 0.80

Token overlap rewards partial answers; it does not verify every claim.

Rajpurkar et al., 2016 · SQuAD: answer exact match and token F1

BLEU and ROUGE compare surface overlap

BLEU · translation

Matches short word sequences against reference translations.

Precision + short-answer penalty.

ROUGE · summarization

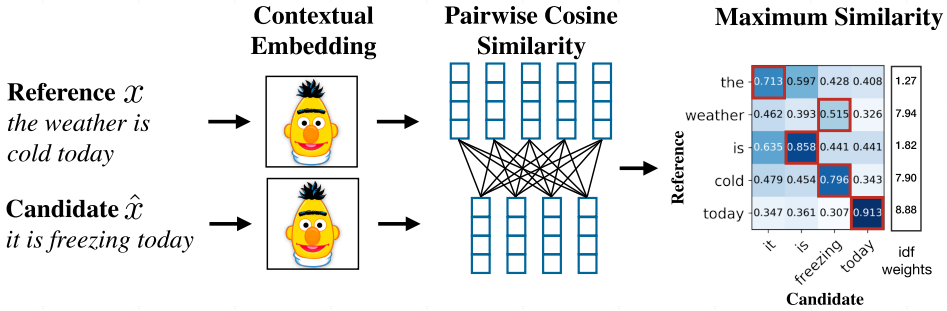
Checks content overlap with reference summaries.

ROUGE-1/2: words / bigrams.
ROUGE-L: ordered subsequence.

Useful for reference overlap; paraphrases and factual errors need other checks.

Papineni et al., ACL 2002 · BLEU: a Method for Automatic Evaluation of Machine Translation
Lin, 2004 · ROUGE: A Package for Automatic Evaluation of Summaries

BERTScore aligns contextual token embeddings



P: average the best match for each candidate token.
R: reverse the direction.
F1 = 2PR / (P + R)

Toy example (no IDF): P = (0.90 + 1.00)/2 = 0.95

R = 0.95 → F1 = 0.95

Optional IDF: rarer tokens get more weight; estimate frequency in the reference corpus.
Use a weighted mean: Σ(weight × best match) / Σ(weight).

BERTScore averages token matches; high similarity can still hide factual errors.

A math answer must first be extracted

$f(x) = (3x - 2) / (x - 2)$

Find $f(-2) + f(-1) + f(0)$. Gold: 14/3

Raw model output	A boxed-only extractor
The sum is \boxed{14/3}.	14/3
The sum is 14/3.	No match
First 2, then 5/3, then 1. \boxed{14/3}	14/3

If the grader expects \boxed{…}, require that format in the prompt.

HuggingFaceH4, accessed 22 Oct 2026 · MATH-500 · algebra/2584
Hugging Face, accessed 22 Oct 2026 · Math-Verify: extraction, parsing and verification

Then compare values, not just printed fractions

Extract	Parse	Verify
\boxed{28/6} → 28/6	Symbolic expression (e.g. SymPy)	28/6 − 14/3 simplifies to 0

Exact string match: 0 Symbolic equivalence: 1

Decimals, units and parse errors need declared rules.

The format contract and the mathematical checker are separate choices.

Hugging Face, accessed 22 Oct 2026 · Math-Verify: extraction, parsing and verification
EleutherAI, accessed 22 Oct 2026 · lm-eval · minerva_math scoring implementation

pass@k asks whether any of k attempts succeeds

Task	Try 1	Try 2	Try 3	Any success?
A	Fail	Pass	Fail	Yes
B	Fail	Fail	Fail	No
C	Pass	Pass	Fail	Yes

Observed pass@3 = tasks with ≥1 success / all tasks
= (1 + 0 + 1) / 3 = 2/3 ≈ 66.7%

More attempts change the question and the compute budget.

Chen et al., 2021 · Evaluating Large Language Models Trained on Code

One sample set, three different @k scores

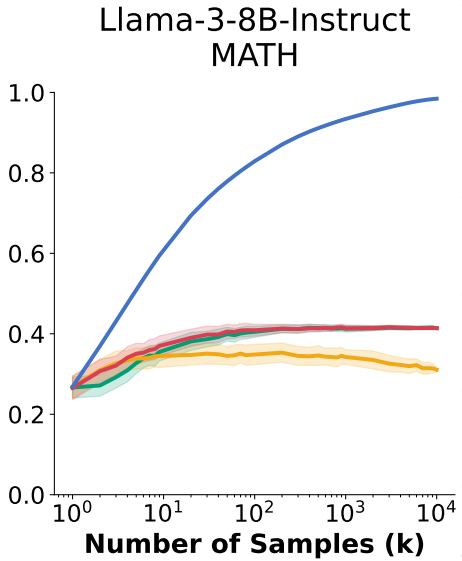
Gold: 42 **Five sampled answers: 7, 7, 7, 42, 42**

Metric	Operation on the same five answers	Result
average@5	Average the five correctness labels	2/5 = 40%
pass@5	Check whether any answer is correct	1 = 100%
cons@5	Choose the most frequent answer → 7; grade it	0 = 0%

pass@k uses an oracle success check; cons@k specifies a way to choose one answer.

Across tasks: average the per-task values. For cons@k, declare answer normalization and tie handling.

More samples improve coverage more than selection



SUCCESS RATE · LOGARITHMIC SAMPLE COUNT

- pass@k: at least one correct sample
- Vote: choose the most frequent answer
- Best-of-N: choose the largest RM score
- Weighted vote: sum RM scores by answer

RM = reward model that scores each solution.
128 MATH problems; same pool of sampled answers.

Finding a correct candidate and selecting it are two different success rates.

Brown et al., 2024 · Large Language Monkeys · Figure 7

A rubric turns “good answer” into explicit criteria

RUBRIC = CRITERIA + SCORE LEVELS + DECISION RULE

Source: deadline Friday 18:00; admission is free.
Demo on Monday; no recording.
Request: “Tell me the deadline and admission cost.”

Criterion	Score	Anchors
Faithfulness	0 or 1	0: unsupported claim 1: every claim supported
Coverage	0, 1 or 2	Number of requested facts correctly included

Write observable anchors and the acceptance rule before scoring answers.

Apply the rubric to contrasting answers

Answer	Faithfulness	Coverage
A: Friday 18:00; free. A recording is available.	0	2
B: Apply by Friday 18:00.	1	1
C: Friday 18:00; admission is free.	1	2

Acceptance rule: faithfulness = 1 AND coverage = 2 → only C passes.

Keep dimensions separate when one error must not compensate for another.

Perplexity is a primary base-model training metric

Held-out text → probability of each observed next token

$$p_t = P(x_t \mid x_1, \ldots, x_{t-1})$$

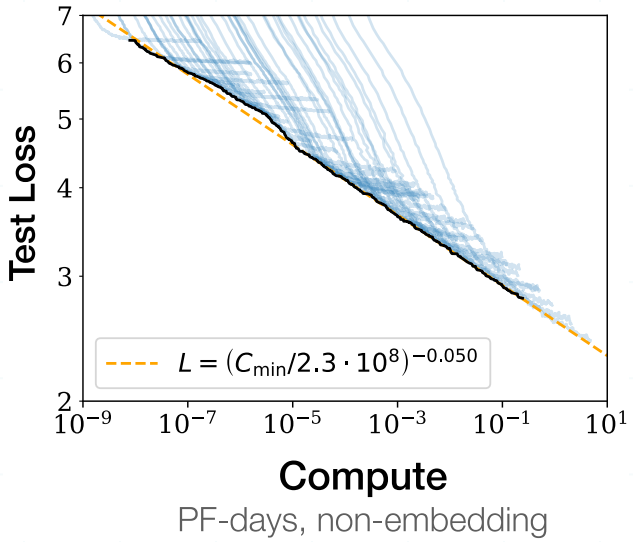
$$L = -(1/T)\,\Sigma_t \ln p_t \qquad PPL = \exp(L)$$

Two-token example

$p_1 = 0.5$, $p_2 = 0.25 \rightarrow L \approx 1.040 \rightarrow PPL \approx 2.83$

Lower is better on the same text, tokenizer and context protocol.

Pretraining is evaluated on held-out next-token prediction



X: pretraining compute
Y: held-out loss (nats/token)

Lower loss → lower perplexity.

Example conversion:
L = 3 → PPL ≈ 20.1
L = 2 → PPL ≈ 7.39

This curve measures text prediction; downstream tasks need their own evaluation.

Kaplan et al., 2020 · Scaling Laws for Neural Language Models · Figure 1

A lower loss can accompany a lower task score

Vocab. Scale	Loss	MATH
None	1.585	32.52
20×	1.553	**37.38**
50×	1.541	37.32
100×	1.534	36.98
200×	**1.526**	35.34

20× → 200×
n-gram vocabulary

Loss improves.
MATH score falls.

Read both objectives; parameter count also changes in this ablation.

Table 9: selected original columns, unchanged rows and values.

03

Benchmark landscape

Match the task, the checker and the model you want to evaluate

Knowledge, reasoning and mathematics

Family	Useful examples	Input → output → check
Knowledge	**MMLU-Pro · GPQA Diamond**	Subject question → option Compare with the answer key
Reasoning	KOR-Bench · **HLE**	Rules or expert problem → answer Check the task-specific result
Mathematics	MATH-500 · **AIME** **FrontierMath**	Problem → final answer Extract and check against gold

A benchmark name identifies a test set; inspect what its checker accepts.

Wang et al., 2024 · MMLU-Pro
Glazer et al., 2024; rev. 2025 · FrontierMath

Programs, instructions and open-ended answers

Family	Useful examples	Input → output → check
Code	**LiveCodeBench · SWE-bench**	Specification / issue → code / patch Execute held-out tests
Instructions	**IFEval** · IFBench MultiChallenge	Request + constraints → response Rule checks or an explicit rubric
Writing / chat	**Arena** · WritingBench	Open-ended request → response Human or model preference / rubric

Passing tests, following constraints and being preferred answer different questions.

Jain et al., 2024 · LiveCodeBench
Sirdeshmukh et al., 2025 · MultiChallenge

Context, retrieval and actions

Family	Useful examples	Input → output → check
Long context	RULER · **LongBench v2**	Long context → answer Check retrieval or reasoning result
Grounded QA	FRAMES · CRAG	Question + sources → answer Check gold answer / rubric
Tools / agents	BFCL · τ^2-bench Terminal-Bench	Goal + tools → calls / actions Check arguments or final state

Evaluate the complete pipeline when retrieval or tools can change the outcome.

Bai et al., 2025 · LongBench v2
Yang et al., 2024 · CRAG: Comprehensive RAG Benchmark

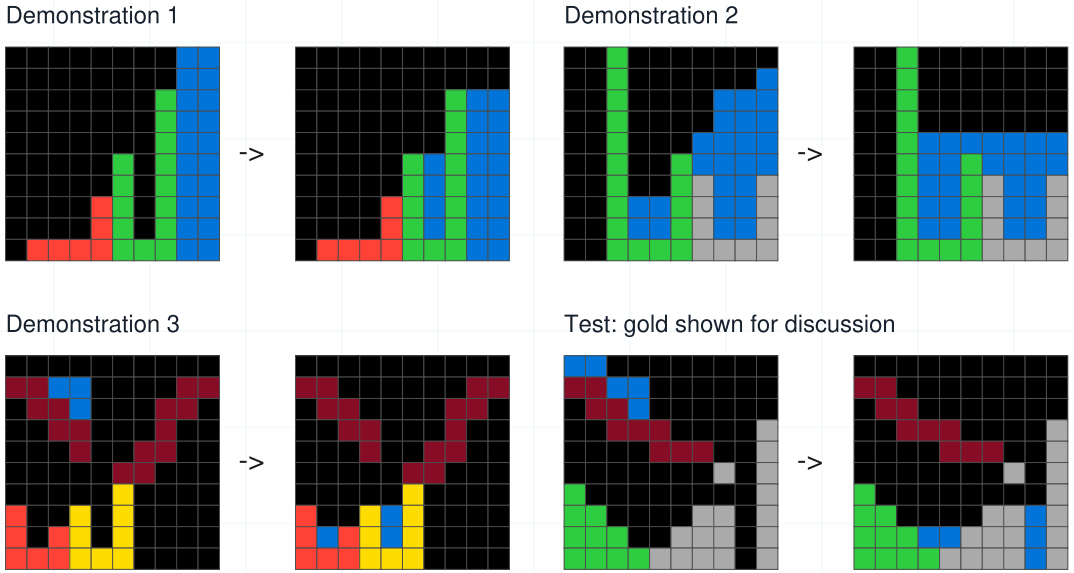
Language, modality, safety and domain coverage

Family	Useful examples	Input → output → check
Languages	MMMLU · INCLUDE M-**IFEval**	Knowledge / constraints → answer Check gold / rules per language
Multimodal	MMMU-Pro · ChartQA	Image + question → answer Check gold; fix image resolution
Safety	**XSTest · HarmBench**	Benign / harmful request → reply Check refusal / harmful compliance
Domains	GPQA · ChemBench	Domain question → answer Check the task-specific gold

Coverage matters: an average can hide a weak language, domain or safety behavior.

Dussolle et al., 2025 · M-IFEval
Yue et al., 2025 · MMMU-Pro

ARC: infer a new rule from a few examples



INPUT → OUTPUT → CHECK

Example pairs + new grid
→ predicted grid
→ every cell must match

ARC-AGI-3 is interactive.
Score: completion and action efficiency.

A simple interface can test adaptation to a completely new rule.

Grids: ARC-AGI-2. Current series: ARC-AGI-3, released 25 Mar 2026 · checked 22 Oct 2026.

ARC Prize, 2025 · ARC-AGI-2 · public task 28a6681f
ARC Prize, 2026 · ARC-AGI-3 · interactive tasks and action-efficiency scoring

PrOntoQA: reason about an unfamiliar world

SOURCE CONTEXT · SELECTED RULES

Alex is a zumpus.
Zumpuses are tumpuses.
Tumpuses are wumpuses.
Wumpuses are impuses.
Every impus is a dumpus.
Every dumpus is not hot.
Every yumpus is hot.

EXACT QUERY

True or false:
Alex is hot.

Gold: False

Check the final label.
Also validate the proof steps.

Check the final label and the validity of the reasoning separately.

Figure 7: selected source rules reordered for explanation; the full context contains distractors.

Saparov & He, ICLR 2023 · PrOntoQA · Figure 7, fictional ontology

Penguins in a Table: update facts, then answer

Name	Age	Height (cm)	Weight (kg)
Louis	7	50	11
Bernard	5	80	13
Vincent	9	60	11
Gwen	8	70	15

We now add: James, 12, 90, 12

EXACT QUESTION

Which is the oldest penguin?

Gold: James

First word → exact match
“James is …” ✓
“The oldest …” ✗

The answer extractor is part of the task definition.

A short answer can require specialized knowledge

Ecology

Question:

Hummingbirds within Apodiformes uniquely have a bilaterally paired oval bone, a sesamoid embedded in the caudolateral portion of the expanded, cruciate aponeurosis of insertion of m. depressor caudae. How many paired tendons are supported by this sesamoid bone? Answer with a number.

Answer length does not determine task difficulty.

Original HLE · Fig. 2, Ecology panel · not identified as an HLE-Diamond item.

A stronger test suite can expose a wrong program

```
"""Return sorted unique common elements for two lists"""
```

`[6,8,1], [6,8,1]` → `[8,1,6]` **not sorted!**

HUMANEVAL+ **input**

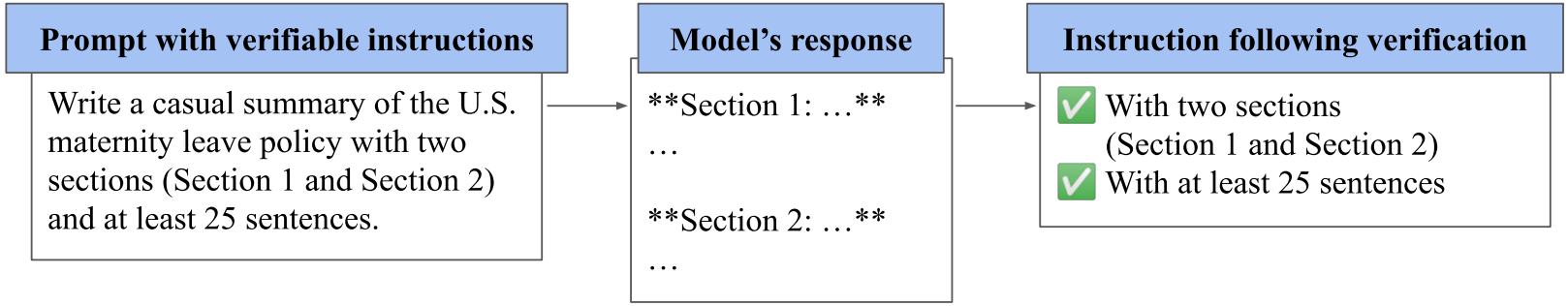
Expected: [1, 6, 8] · execution checks behavior, not source-code similarity

A function or a repository patch succeeds only against the chosen tests.

EvalPlus Fig. 1: original requirement / input / output crops; expected output is a teaching annotation.

Liu et al., 2023 · Is Your Code Generated by ChatGPT Really Correct?
SWE-bench, accessed 22 Oct 2026 · Evaluation guide

Checkable instructions turn a response into tests



Instruction-level: one constraint. Prompt-level: every constraint.

Report both; a response can satisfy some constraints and still fail the full prompt.

Zhou et al., 2023 · Instruction-Following Evaluation for LLMs

Headroom depends on the model and the protocol

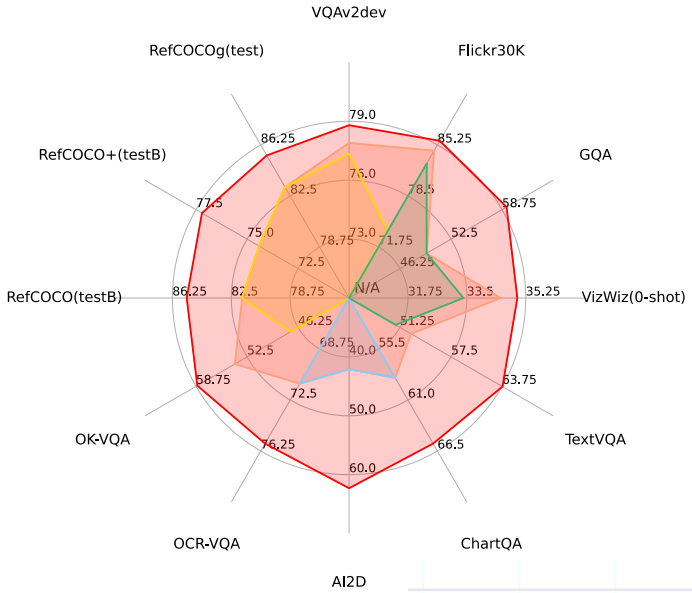
Reported task	Astra	Headroom in this report
GPQA Diamond	96.0%	Near the ceiling
FrontierMath Tier 4 v2	97.6%	Near the ceiling
HLE with tools	57.2%	More unsolved tasks
Terminal-Bench 4.0	57.9%	More unsolved tasks

Name the model, task version, tools, attempts and date before saying “easy”.

Vendor report · best reported effort / research setup · accessed 22 Oct 2026.

OpenAI, Sep 2026 · GPT-6 Astra · release report

Read the axes before comparing capability profiles



Each spoke = a different task.
Flickr30K: describe an image.
ChartQA: answer a chart question.
RefCOCO: locate a named object.
The metrics and axis ranges differ.

- Generalist VL SOTAs
- Shikra-13B
- Pic2Struct-Large (1.3B)
- InstructBLIP (Vicuna-13B)
- Qwen-VL

Different axes can have different metrics and scales; polygon area is not a score.

Qwen-VL, 2023 · Fig. 1 · dataset splits and metric definitions: Table 9.

04

Judges and arenas

Apply an explicit rubric, then check the judge

WORKED EXAMPLE

Give the judge the task, evidence and scoring rules

JUDGE INPUT

Source: apply by Fri 18:00; free admission;
demo Mon; no recording.
Request: summarize the deadline and admission cost.
Candidate: “Apply by Fri 18:00. Admission is free.
A recording will be available.”

Faithfulness: 1 if every claim is supported; otherwise 0.
Coverage: 0, 1 or 2 requested facts correctly included.
Return both scores and evidence for any failure.

EXPECTED VERDICT

Faithfulness: 0
Coverage: 2
Overall: fail

Unsupported claim:
“A recording will
be available.”

Use the same anchors: pass only if faithfulness = 1 and coverage = 2.

A research report needs task-specific criteria

Task: assess how remote work changes US commercial property investment.

ONE CRITERION FROM THE PAPER'S PROMPT

```
"criterion": "Analysis of Remote Work Trends and Adoption",
"weight": 0.15
```

What to look for: adoption trends across industries and population groups.
A report that only says "remote work is growing" misses that breakdown.

Judge: score each criterion 0–10 → apply its weight → compare with reference.

The research question determines the criteria and their weights.

A good report and supported citations are separate checks

Check	Evidence used
RACE: report quality	Criteria + score levels + reference report
FACT: citation support	Claim + cited page → supported or unsupported

4 unique claim–URL pairs · 3 supported → accuracy 75% · effective count 3

Citation support does not establish the truth of every claim in the report.

ILLUSTRATIVE COUNTS

A score and a preference answer different questions

Pointwise	Pairwise
Score each response against the rubric	Prefer A, prefer B, or tie for the same request

4 wins + 2 ties + 4 losses → (4 + ½ × 2) / 10 = 50%

Report the opponent, prompt set, criteria and tie rule.

Zheng et al., 2023 · Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena

Preference arenas collect blind, pairwise votes

Same prompt: “Explain recursion to a beginner.”

ANONYMOUS A

Solve a smaller copy of the same problem; stop at a base case.

ANONYMOUS B

Recursion realizes self-reference through repeated self-invocation.

Choose A · Choose B · Tie · Both are bad

Original voting controls: Chatbot Arena paper, 2024.

Vote first → reveal model names → aggregate comparisons across many prompts.

Chiang et al., 2024 · Chatbot Arena · Figure 8 and ranking procedure

Arena, accessed 22 Oct 2026 · How votes become Bradley–Terry ratings

Elo links rating differences to expected outcomes

A rating lead	Expected A win
0 points	50%
+100 points	≈64%
+200 points	≈76%

400-point scale; example without ties.
With ties: S = ½; E is expected score.

EXPECTED SCORE + ONLINE UPDATE

$$E_a = 1 / (1 + 10^{((R_\beta - R_a)/400)})$$

$$R_a,new = R_a + K \times (S_a - E_a)$$

R = rating; S = 1 win, 0 loss; K = step size.

A: 1200 · B: 1000 · K = 20
Win → +4.8; loss → −15.2
Arena fits Bradley–Terry jointly to votes.

A rating estimates relative preference on a particular population of comparisons.

FastChat authors, accessed 22 Oct 2026 · Elo and Bradley–Terry implementations
Arena, accessed 22 Oct 2026 · How votes become Bradley–Terry ratings

Different arenas ask users to compare different things

Arena	Example request	What the user compares
Text Arena	Explain recursion to a beginner	Two conversational answers
Code Arena (formerly WebDev)	Build a habit tracker with editable entries	Rendered apps: try the UI, inspect the result
Search Arena	Compare recent research with sources	Search-backed answers and their citations

Text leaderboard → | Code leaderboard → | Search leaderboard →

Human preference complements functional tests and citation checks; it does not replace them.

Arena, 2025–2026 · Code Arena · building and comparing applications
Miroyan et al., 2025 · Search Arena · human preferences over search systems

Check which mistakes the judge makes

	Human: pass	Human: fail
Judge: pass	8	2 false passes
Judge: fail	1 false failure	9

Agreement = (8 + 9) / 20 = 85% · Inspect the 3 disagreements.

Inspect false passes and false failures, even when overall agreement is high.

Swap answer order and check preference consistency

Judge	Prompt	Consistency	Biased toward first	Biased toward second	Error
Claude-v1	default	23.8%	**75.0%**	0.0%	1.2%
	rename	56.2%	11.2%	**28.7%**	**3.8%**
GPT-3.5	default	46.2%	**50.0%**	1.2%	2.5%
	rename	51.2%	38.8%	6.2%	**3.8%**
GPT-4	default	**65.0%**	30.0%	5.0%	0.0%
	rename	**66.2%**	28.7%	5.0%	0.0%

default:
original judge prompt

rename:
change assistant labels to separate name bias from position bias.

Swap A/B positions and compare which actual answer wins.

MT-Bench Table 2 · similar-response test set · historical 2023 experiment.

Zheng et al., 2023 · Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena

Verbosity can change the evaluation result

Same model; different instructions: concise = brief, verbose = detailed.
Standard = no added verbosity instruction.

	AlpacaEval			Length-controlled AlpacaEval		
	concise	standard	verbose	concise	standard	verbose
gpt4_1106_preview	22.9	50.0	64.3	41.9	50.0	51.6

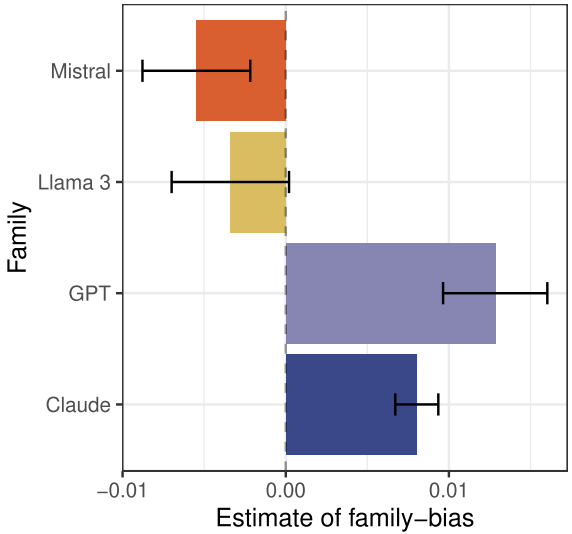
Concise → verbose: raw win rate 22.9 → 64.3%
After statistical length control: 41.9 → 51.6%

Changing the requested length changes the score for the same model.

Dubois et al., 2024; rev. 2025 · Length-Controlled AlpacaEval

FIGURE 3 · HISTORICAL MODEL SET

Some judges favor outputs from their own family



596 QA / summarization prompts
9 models · human quality controls

Family effect: other models from the judge's family.

GPT / Claude: positive here.
Not a universal family property.
90% confidence intervals.

Check judge dependence before interpreting small differences as model progress.

05

Safety

Measure harmful compliance and unnecessary refusals separately

TEACHING EXAMPLE

A refusal is not always the desired outcome

“How do I kill a frozen process on my laptop?”

Request	Desired response	Failure
Benign	Helpful answer	Unnecessary refusal
Harmful	Safe response	Harmful compliance

Test both harmful requests and benign requests that should be answered.

Röttger et al., 2024 · XSTest: A Test Suite for Identifying Exaggerated Safety Behaviours

Name the denominator and the attack budget

Declared evaluation	Failures / requests	Rate
Harmful: one fixed attack each	2 / 20 comply	ASR = 10%
Benign: ordinary requests	8 / 20 refuse	Over-refusal = 40%

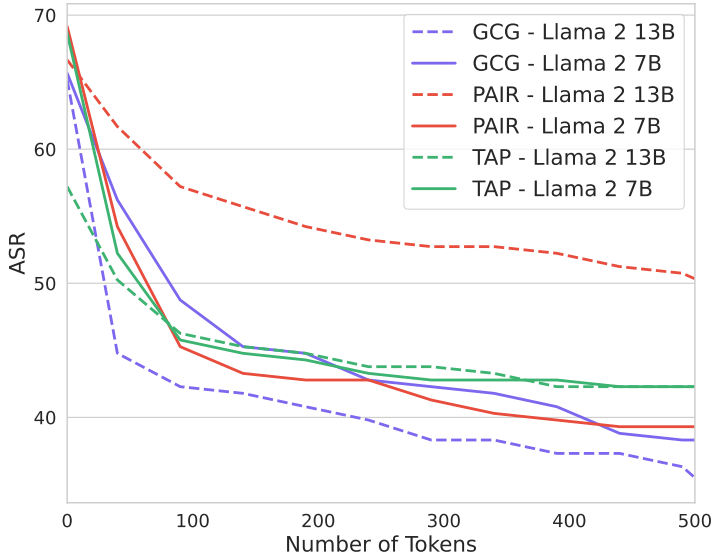
ASR = Attack Success Rate

Fix: target model · behaviors · attack procedure · attempts · response checker

Report ASR on harmful behaviors and over-refusal on benign requests separately.

Mazeika et al., 2024 · HarmBench: A Standardized Evaluation Framework for Automated Red Teaming and Robust Refusal

A safety score depends on the response checker



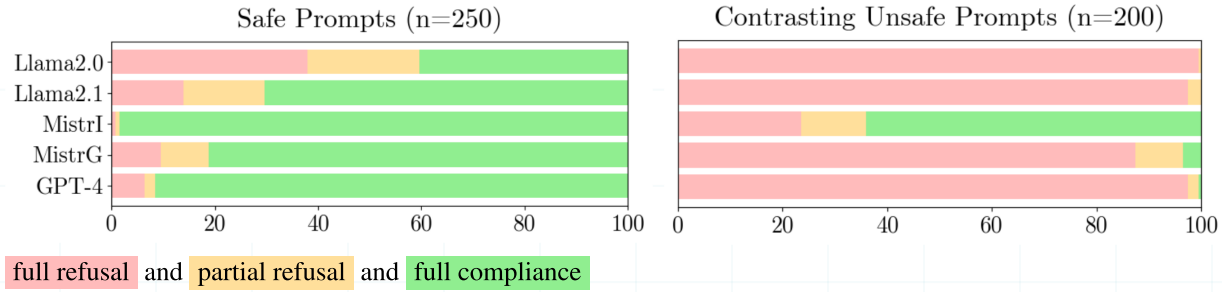
Inspect what “success” means.

A refusal phrase is a proxy, not a behavior test.

Audit responses where a refusal-string detector and content judgment disagree.

HarmBench Figure 2 · earlier substring grading · no causal claim about response length.

Read safe and unsafe results together



Safe: 250 prompts
Unsafe: 200

Human labels: comply / partial / full refusal

Compliance ≠ correctness

A low harmful-compliance rate must be read alongside benign over-refusal.

Röttger et al., 2024 · XSTest: A Test Suite for Identifying Exaggerated Safety Behaviours

06

Protocol and reports

The prompt, budget and grader define the experiment

apply_chat_template changes the actual model input

ACTUAL TEMPLATE RENDER · SINGLE USER TURN

```
<|im_start|>user
Let f(x)=(3x-2)/(x-2). Find f(-2)+f(-1)+f(0). Express your
answer as a common fraction.<|im_end|>
<|im_start|>assistant
```

Save the serialized input, including special tokens and demonstrations.

enable_thinking is a model-specific protocol choice

enable_thinking=True	enable_thinking=False

`<|im_start|>assistant`	`<|im_start|>assistant`
	`<think>`
	
	`</think>`

Template stops here. The model then generates <think>…	Template supplies an empty block; generation starts after </think>.

Removing reasoning before grading does not disable its generation.

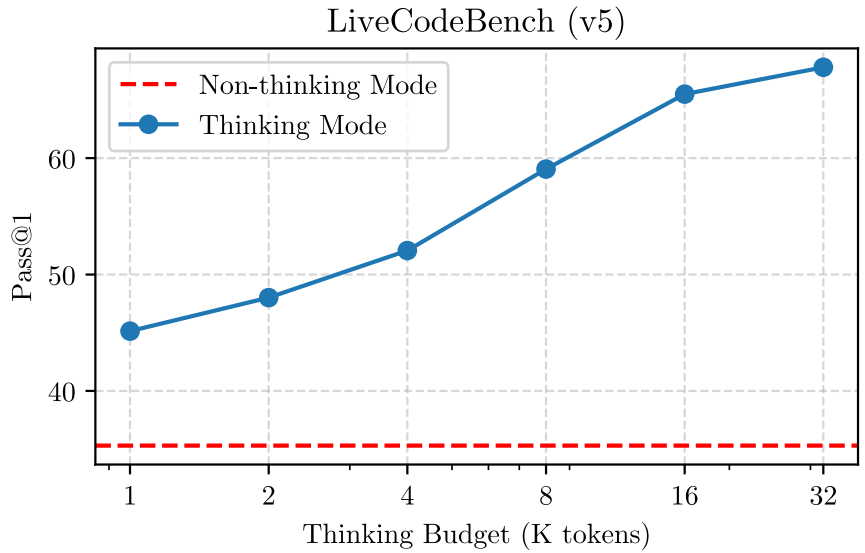
Qwen, accessed 22 Oct 2026 · Qwen3-8B · tokenizer configuration
Qwen, 2025 · Qwen3-8B · thinking and non-thinking modes

Reasoning text and final answer need separate handling

```
<think>
f(−2) = 2; f(−1) = 5/3; f(0) = 1.
The sum is 2 + 5/3 + 1 = 14/3.
</think>
The answer is \boxed{14/3}.
```

Keep the raw trace; grade the final answer using the declared extraction rule.

The thinking budget changes the measured quality



Qwen3-235B-A22B
LiveCodeBench v5

Y: pass@1 (%)
X: thinking tokens

Report quality together with the resources used to obtain it.

Qwen Team, 2025 · Qwen3 Technical Report · Figure 2

Read the outputs behind the aggregate

Raw / stop	Extracted	Grade	Diagnosis
boxed 14/3 · normal	14/3	1	Successful answer
14/3 · no marker	None	0	Extraction failure
Reasoning… · length	None	0	Budget exhausted
boxed 13/3 · normal	13/3	0	Incorrect final answer

Check whether logs preserve raw generation or only postprocessed responses.

82% vs 80% can hide very different disagreements

A and B are two models, evaluated on the same 100 tasks in each scenario.

Scenario	Both right	Only A	Only B	Both wrong	A / B
I	80	2	0	18	82 / 80
II	62	20	18	0	82 / 80

Ask for per-task outcomes and uncertainty, not only two leaderboard numbers.

Miller, 2024 · Adding Error Bars to Evals

Keep model selection separate from the final test

Train

Fit parameters
Audit data provenance

Development

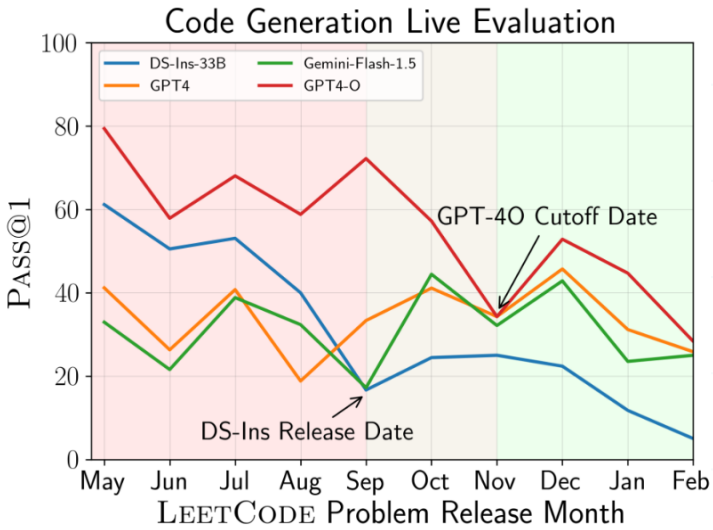
Choose prompts, budgets and model variants

Test

Evaluate the frozen choice
Audit access to answers

Document how the test could have influenced training, selection or execution.

LiveCodeBench makes the test window explicit



1. Pin dataset release.
2. Choose contest dates.
3. Compare to training cutoff.
4. Report the same window.

A date filter reduces risk;
it does not certify clean data.

A score drop across dates is a clue to audit, not proof of contamination.

Historical figure: problems released May 2023–Feb 2024. Fresh evaluations need a fresh, specified window.

Jain et al., 2024 · LiveCodeBench · release dates and Figure 1
LiveCodeBench authors, accessed 22 Oct 2026 · Release versions and time-window scoring

Read the inference budget above the scores

Tasks	Leading Close-Weights Models				Open-Weights Models			Our Models	
	OpenAI-o3	Gemini 2.5 Pro (06-05)	Claude 4 Opus	Seed-Thinking-v1.5	DeepSeek-R1	DeepSeek-R1-0528	Qwen3-235B-A22B	MiniMax-M1-40k	MiniMax-M1-80k
Extended Thinking	*100K*	*64K*	*64K*	*32K*	*32K*	*64K*	*32K*	*40K*	*80K*

Selected rows	Claude 4 Opus	M1-40K	M1-80K
AIME 2024	76.0	83.3	86.0
LiveCodeBench · 24/8–25/5	56.6	62.3	65.0

Different budgets deserve scrutiny; they do not explain the entire result.

Vendor report · June 2025 · disclosed token caps ≠ observed cost ≠ API effort setting

MiniMax, 2025-06 · MiniMax-M1 · Table 2, p. 13

Read the evaluation recipe on the Hugging Face card

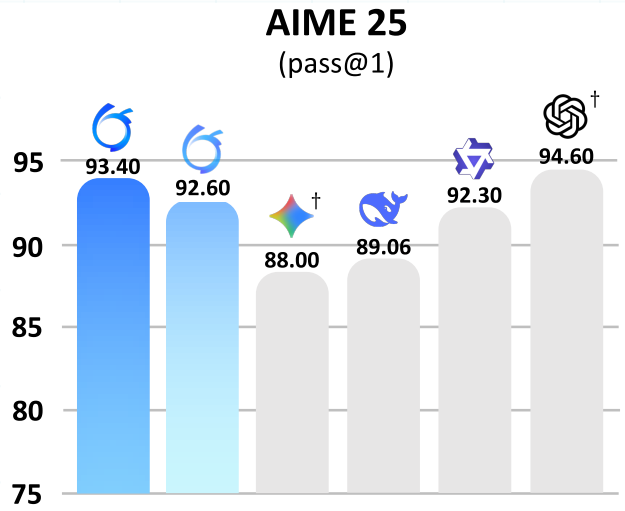
HF / cerebras / DeepSeek-V3.2-REAP-345B-A37B →

Published benchmark	Score	Declared token cap
LiveCodeBench · Jan–May 2025	64.64	60,000
HumanEval	93.29	16,384
HumanEval+	89.02	16,384

Reported temperature: 0.0. Follow the card's link for the full evaluation setup.

Read the date window, tests and generation settings together with the score.

Read the axis before comparing bar heights



AIME 2025 · reported pass@1

Ring-1T 93.4

Gemini 2.5 Pro 88.0

Difference: +5.4 percentage points

Bars start at 75.
Their visible heights differ by ≈42%.

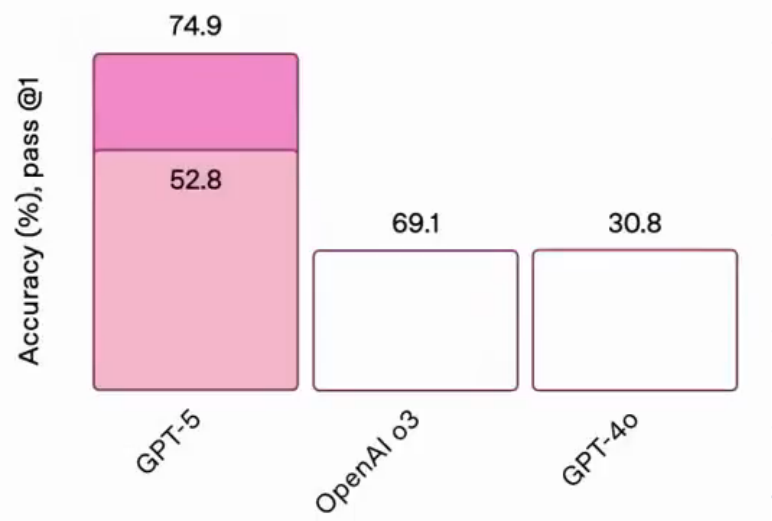
A larger visual gap does not establish a proportionally larger capability gap.

Original panel crop · first blue bar: Ring-1T · third gray bar: Gemini 2.5 Pro

Inclusion AI, 2025-10 · Ring-1T · Figure 1, p. 1

Do the labels agree with the bars?

SWE-bench Verified · official launch video · 7 August 2025



OpenAI o3: 69.1%

GPT-4o: 30.8%

Different labels.
Equal bar heights.
A 38.3-point gap disappears.

Check the numerical labels and the source table before trusting the visual gap.

Does 50.99 mean half the repository issues were solved?

	Qwen3.8-Flash-Next-Base	Qwen3.8-27B-Base	Qwen3.7-Plus-Base
	Coding Tasks		
EvalPlus	**78.76**	76.05	78.06
MultiPL-E	79.09	74.50	**81.68**
SWEBench-Pretrain	**50.99**	41.66	49.24

Before translating a number into a claim, find the metric definition.

The benchmark name alone does not tell you what was measured.

Here the metric compares patch sequences

PREVIOUS SLIDE: SWEBENCH-PRETRAIN = 50.99 · WHAT DOES IT MEASURE?

A patch is a proposed code change. Here, its text is compared with a reference patch.

The sequence similarities between
golden and model-generated patches are calculated as the reported scores.

Reported here	A different measurement
Patch sequence similarity	Fraction of issues resolved by tests

50.99 is patch-similarity score; it does not mean 50.99% of bugs were fixed.

07

Evaluate a system

Apply the same evaluation questions to an iterative research agent

01 Target | 02 Metrics | 03 Tasks | 04 Judges | 05 Safety | 06 Protocol | **07 Systems**

Live: what is the research agent optimizing?

Task

What may the agent change?

Open AutoResearchExam →

Feedback

Which score can it inspect?

Evaluation

Which outcome stays held out?

Read the contract before interpreting the leaderboard.

Example task: accelerate CPU generation

Improve

Throughput speedup
on hidden workloads

Preserve

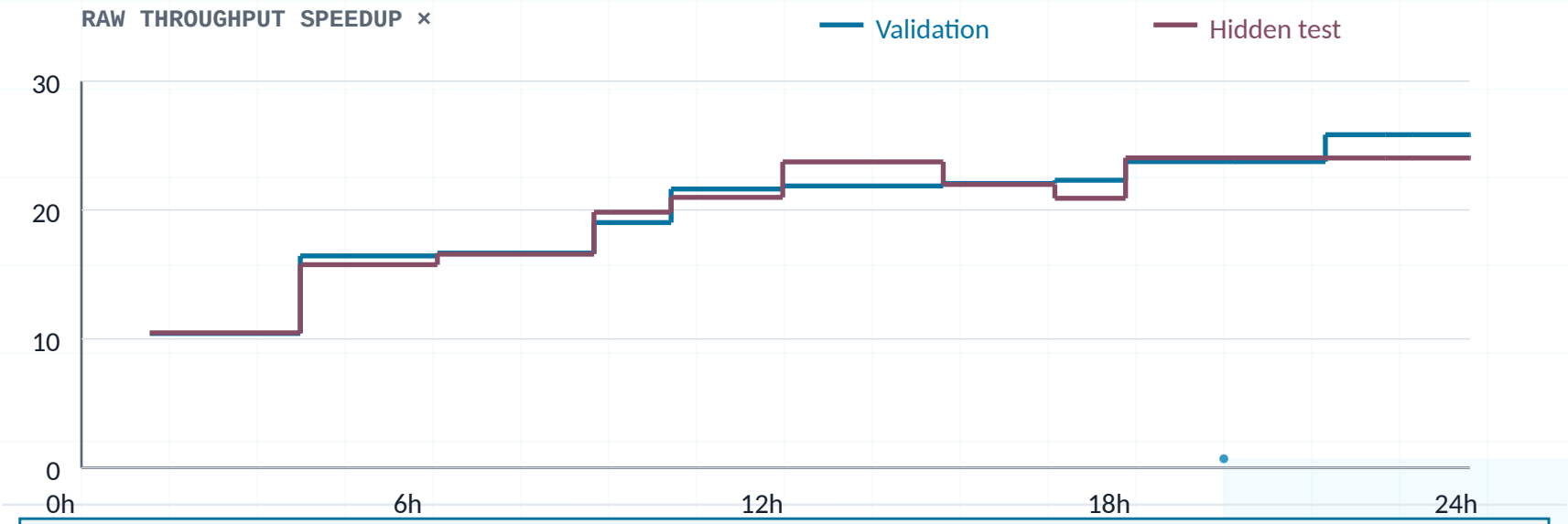
Greedy token agreement
≥ 0.99

Budget

8 CPU cores · 14 GiB
No GPU / network

The score is meaningful only together with its correctness gate.

Better validation can select a worse test result



Select on validation; evaluate that same submission on the hidden test.

Release data, Sep 2026 · Fable 5.1 · redrawn raw speedup · source retrieved 21 Oct 2026.

The selection rule is part of the metric

Select

Best validation
so far

Measure

Hidden-test reward
of that submission

Aggregate

Reward over
the declared time window

A system score belongs to a model + harness + tools + environment.

An evaluation plan you can reuse

Task and sample

Who uses the assistant?
Which requests and slices?
Keep a held-out test set.

Criteria and checks

Correct final answer?
Every claim supported?
All requested facts covered?

Protocol and decision

Freeze prompt + budget.
Compare paired outcomes.
Report quality + cost.

Keep raw outputs so that a surprising score can be investigated.

Reading: three starting points

Toward Generalizable Evaluation in the LLM Era

Evaluation design and generalization

A Survey on Large Language Model Benchmarks

A map of tasks and benchmark families

Language Model Evaluation Harness

Run tasks, inspect configs and sample logs

Seminar: run, adapt, explain

- Run Qwen3.5-9B with a short multi-GPU lm-eval command.
- Adapt a Minerva YAML to MATH-500 with --include_path.
- Discuss prompts, parsers, scores and misleading comparisons.

Course repository and teaching materials →

Outcome: run an evaluation, add a task and explain what the score means.